

Effects of Temperature on the Mortality and Growth of Hawaiian Reef Corals*

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Abstract

Three common species of Hawaiian reef corals, *Pocillopora damicornis* (L.), *Montipora verrucosa* (Lamarck) and *Fungia scutaria* Lamarck, were grown in a temperature-regulated, continuous-flow sea water system. The skeletal growth optimum occurred near 26°C, coinciding with the natural summer ambient temperature in Hawaii, and was lowest at 21° to 22°C, representing Hawaiian winter ambient. Levels of approximately 32°C produced mortality within days. Prolonged exposure to temperatures of approximately 30°C eventually caused loss of photosynthetic pigment, increased mortality, and reduced calcification. Corals lived only 1 to 2 weeks at 18°C. The corals showed greater initial resistance at the lower lethal limit, but ultimately low temperature was more deleterious than high temperature. Results suggest that a decrease in the natural water temperature of Hawaiian reefs would be more harmful to corals than a temperature increase of the same magnitude.

Introduction

The response of reef corals to short-term (minutes to hours) incubations at various temperatures have been described by Mayer (1918), Edmondson (1928), Yamazato (1970) and Clausen and Roth (1975). Long-term temperature effects on reef coral growth and mortality have been studied only in the field (Shinn, 1966; Maragos, 1972; Glynn and Stewart, 1973; Jokiel and Coles, 1974). These results are often influenced by local variations in factors such as turbidity, sedimentation, light, water motion, and salinity. Investigators have also used zoogeographic data to define temperature optima and limits for corals and coral reefs (Vaughan and Wells, 1943; Wells, 1957; Ma, 1958; Rosen, 1971; Stehli and Wells, 1971). Again, numerous important environmental factors other than temperature vary regionally, complicating interpretation of these data. The present investigation was undertaken to reduce these complications by using controlled laboratory experiments to investigate the long-term effects of temperature on growth and mortality of Hawaiian corals.

Materials and Methods

Experiments were conducted in fiberglass aquaria located outdoors in full natural light. Light intensity at the bottom of the 0.5 m deep aquaria is therefore nearly identical to light at that depth in the field. The corals used in the experiments were removed from shallow Kaneohe Bay reef flats which have a mean depth approximating the depth of the aquaria. Temperature-regulated, non-recirculating, unfiltered sea water was pumped from Kaneohe Bay, Oahu, Hawaii, at a rate sufficient to flush the aquaria in less than 1 h. Only inert materials contacted the sea water: pump and piping surfaces were of non-toxic plastic; heat exchange surfaces were of titanium. Temperatures in each aquarium were recorded for 5 min periods at hourly intervals with a scanning thermistor telethermometer and recorder accurate to $\pm 0.1^\circ\text{C}$. Dissolved oxygen, bacteria concentration, pH, plankton and other conditions were measured in the tanks but did not depart from field conditions because of the high flushing rate. Corals in the tanks at ambient temperature grew at the same rate as corals measured simultaneously in similar shallow field environments (unpublished data and Maragos, 1972). Four temperature treatments were used in each experiment.

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The effect of temperature on coral skeletal growth rate and mortality was measured for 3 species: *Pocillopora damicornis* (L.), a highly branched hemispherical coral which is representative of colonial species having a fixed upper size; *Montipora verrucosa* (Lamarck), a fast-growing acroporid coral which can become quite large and can assume encrusting, massive foliaceous or branching growth forms; *Fungia scutaria* Lamarck, a solitary coral which reaches a maximum size of several hundred grams wet weight.

In the first three experiments, 60 corals (20 of each species) were used in each of 4 treatments. Unattached hemispherical colonies of these species were collected from the reef flat adjacent to the laboratory. Each colony was gently lifted off the substratum and placed in a submerged container, transported to the aquaria, submerged, and placed in its final position. The corals were not exposed to drying or abrasion, and we saw no evidence of tissue damage. These corals ranged in weight from several grams to over 200 g wet weight, allowing evaluation of the effect of initial size on growth and mortality. The corals were randomly assigned to positions in a matrix of 6 rows of 10 columns. Skeletal growth was measured using a buoyant weight method (Jokiel *et al.*, in press).

Experiment 1 was conducted during the summer and was designed to measure growth in the 27° to 31°C range and to establish upper temperature limits. Each of the groups of corals was set up in an identical manner, acclimated to tank conditions at ambient summer Kaneohe Bay temperature (26° to 27°C) for a 30-day period, and weighed. One tank was held at ambient temperature, while the water temperatures of the remaining three tanks were increased at a rate not exceeding 2°C per day to final temperatures averaging approximately 28°, 29°, and 31°C. After 30 days of treatment, the corals were reweighed.

Experiment 2 was run the following winter and designed to test growth in the 18° to 26°C range and to establish the lower lethal limit. The experiment was set up as in Experiment 1, with 30 day acclimation at a mean ambient temperature of 23.4°C. After the corals were weighed, temperatures were slowly changed to final means of approximately 18°, 23°, 25°, and 26°C. The corals were reweighed after 25 days of treatment.

Experiment 3 was performed in spring, and measured growth at various temperatures and the ability of Hawaiian corals to adapt to upper sublethal temperatures. Temperature treatments used were approximately 23°, 25°, 27°, and 28°C for the

first 30 day growth period followed by 30 days at approximately 25°, 26°, 27°, and 28°C.

Experiment 4 was conducted a year later in early spring, and measured coral growth for *Pocillopora damicornis* across a range from lower to upper sublethal temperatures. Eight large heads of this species were acclimated and grown for 30 days in each of 4 temperature treatments (20°, 23°, 25°, 29°C), and their weight increases determined.

The sample sizes used in Experiments 1-4 were too small to determine significant differences among mean growth rates for the various temperature treatments. Therefore, a larger sample of a single species (*Montipora verrucosa*) was used in Experiment 5. Sixty colonies were grown for almost 2 months (late spring) at approximately the same 4 temperature regimes used in Experiment 4.

In Experiment 6 the relative thermal resistance of mature colonies of *Pocillopora damicornis* versus new settlements was tested. Eight large colonies weighing several hundred grams each were grown for 60 days at a temperature which had been previously determined to be optimal (26°C). Reproduction occurred during this period, and 59 new colonies (1 to 23 polyps per colony) were observed on a removable standpipe in the tank. The temperature of the tank was increased to 30°-32°C for a period of 7 days before being returned to ambient temperature (25° to 26°C). Condition of the mature colonies and new settlements was compared.

Corals subjected to high temperatures lose their symbiont pigmentation, a condition that has been referred to as "bleached" (Yonge and Nicholls, 1931; Jokiel and Coles, 1974). Bleached corals held for several months at 30°C were returned to a depth of approximately 1 m on the reef (ambient temperature approximately 26°C) and observed for recovery. Normal pigmented corals held in ambient temperature experimental tanks for the same period of time were set out with the bleached specimens as controls. Ten specimens each of *Pocillopora damicornis*, *Montipora verrucosa*, and *Fungia scutaria* along with normally pigmented controls were placed in areas of living coral and observed for a period of 70 days.

Results

Mortality

The results of all temperature treatments of the first three experiments are summarized in Table 1. The warmest treat-

ment in Experiment 1 was lethal to all *Montipora verrucosa*, to all but 1 *Pocillopora damicornis*, and to all but 8 *Fungia scutaria* (Fig. 1). When the temperature reached 32°C, coral polyps contracted. After 24 h, some of the polyps were again expanded but the corals were in the process of losing zooxanthellar pigment. Coral polyps darkened as coloration faded in the coenosarc tissue, apparently due to migration of the zooxanthellae. Within 2 days the polyps also became pale. During this period, temperature readings for the upper lethal treatment held in the 31° to 32°C range, with slight spikes above this value to 33°C at noon and occasional drops during the night to 30°C. Initial mortality was high (Fig. 1), and the pale survivors continued to lose pigment until the living polyps became colorless. On the 13th day of treatment, the surviving colonies (8 *Fungia scutaria* and 1 *Pocillopora damicornis*) were exposed to a 7 h period of over 33°C followed by 3 h at 34.6°C, after which temperatures resumed the 31° to 32°C range. The corals were able to survive, suggesting acclimation to higher temperatures. Although rapid mortality occurred in Experiment 1 only at the highest (summer ambient +4°C) temperature treatment, sublethal effects (loss of pigment) were noted after several weeks in the 29.6°C (ambient +3°C) treatment.

In Experiment 2, lethal effects were observed only in the coldest treatment (18.3°C) (Fig. 1). During the first few days of treatment, the corals appeared normal, but by the 8th day of treatment over half of the *Pocillopora damicornis* specimens were losing tissue. By the 25th day of treatment, nearly all of surviving specimens were in very poor condition, having gradually lost much tissue. At this time the corals were weighed and the temperature returned to ambient, but mortality continued to occur. Only one coral (*Fungia scutaria*) survived the treatment.

Corals held at upper sublethal temperatures in Experiment 3 responded in a manner similar to corals held in the sublethal (29.6°C) treatment during Experiment 1. At temperatures approaching 30°C, the corals appeared to remain healthy for several weeks but then began to lose pigment, becoming pale and then bleached followed by high mortality rates. Table 2 summarizes observations after 30 and 60 days of treatment. The results indicate thermal effects to have been continuously deleterious in the upper sublethal treatment. As in Experiments 1 and 2, *Pocillopora damicornis* appeared to be more sensitive than *Montipora verrucosa* and *Fungia scutaria*. We did not observe

Table 1. Coral mortality for Experiments 1-3. SD: standard deviation

Experiment	Mean temperature ± SD (°C)	% Mortality		
		<i>Pocillopora damicornis</i>	<i>Montipora verrucosa</i>	<i>Fungia scutaria</i>
1	31.0±1.17	95	100	60
	29.6±1.08	0	0	0
	28.5±1.11	0	0	0
	26.7±0.81 ^a	0	0	0
2	26.1±0.61	0	0	0
	24.9±0.74	0	0	0
	23.3±0.69 ^a	0	0	0
	18.3±0.64	100	100	95
3 (first 30 days)	28.3±1.30	5	0	5
	26.8±1.14	0	0	0
	25.0±1.07 ^a	5 ^b	5 ^b	5 ^b
	22.9±1.32	0	0	0
3 (second 30 days)	28.4±1.18	40	10	5
	27.4±1.15	0	0	0
	26.1±0.97 ^a	0	0	0
	25.0±0.75	0	0	0

^aAmbient treatment.

^bMortality related to interruption of waterflow.

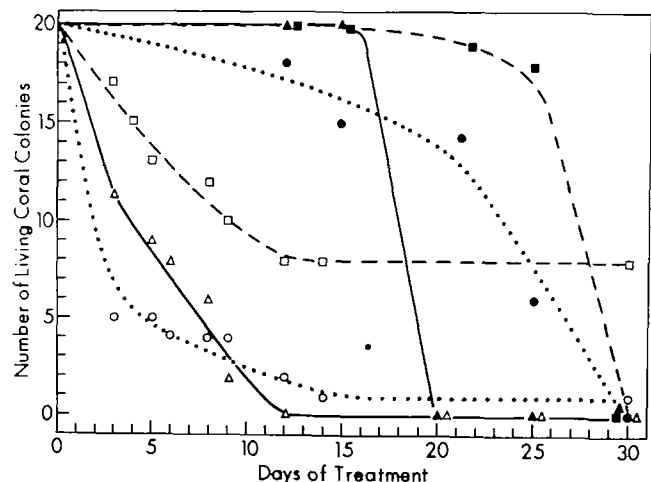


Fig. 1. Mortality rates of 3 Hawaiian reef corals at upper (open symbols) and lower (solid symbols) lethal temperatures. *Pocillopora damicornis*: circles, dotted lines; *Montipora verrucosa*: triangles, solid lines; *Fungia scutaria*: squares, dashed lines

any relationship between initial size (2 to 200 g wet weight) and temperature tolerance.

When corals were exposed to abnormally high or low temperatures, the first evidence of pigment and tissue loss always occurred on the upper surfaces of the corals, which were exposed to full

Table 2. Condition of corals during exposure to sublethal temperature (29° to 30°C) for 20 specimens of each of 3 coral species in Experiment 3

Species	Days of Treatment	Condition			
		Normal	Pale	Bleached	Dead
<i>Pocillopora damicornis</i>	0	20			
	30	4	12	3	1
	60	0	3	9	8
<i>Montipora verrucosa</i>	0	20			
	30	7	11	2	0
	60	3	7	8	2
<i>Fungia scutaria</i>	0	20			
	30	6	10	3	1
	60	2	7	10	1

Table 3. *Montipora verrucosa*. Effect of temperature on growth (Experiment 5). Coral growth at different temperatures (means and standard errors) expressed as $\Delta\sqrt{\text{weight}}$. Pooled SE = 0.00714; D = 0.0259

Temperature (°C)	$\Delta\sqrt{\text{weight}}$		Mean differences			
	Mean	SD	$\bar{X}$	SE	n	28.1 25.8 23.6 19.8
28.1	1.2	.103	.0067	55		.029 ^a .017 -.059 ^a
25.8	0.9	.132	.0095	59		-.012 -.088 ^a
23.6	1.1	.120	.0071	57		-.076 ^a
19.8	1.2	.044	.0040	59		

^aSignificant at the 0.05 level.

Table 4. Recovery of bleached corals returned to Coconut Island reef

Species	Day of observation	Condition			
		Normal	Pale	Bleached	Dead
<i>Pocillopora damicornis</i>	0			10	
	14		10		
	39		8		2
	70		7		2
<i>Montipora verrucosa</i>	0			10	
	14		5	3	2
	39		8		2
	70	7			3
<i>Fungia scutaria</i>	0			10	
	14		9		1
	39		8		2
	70	8			2

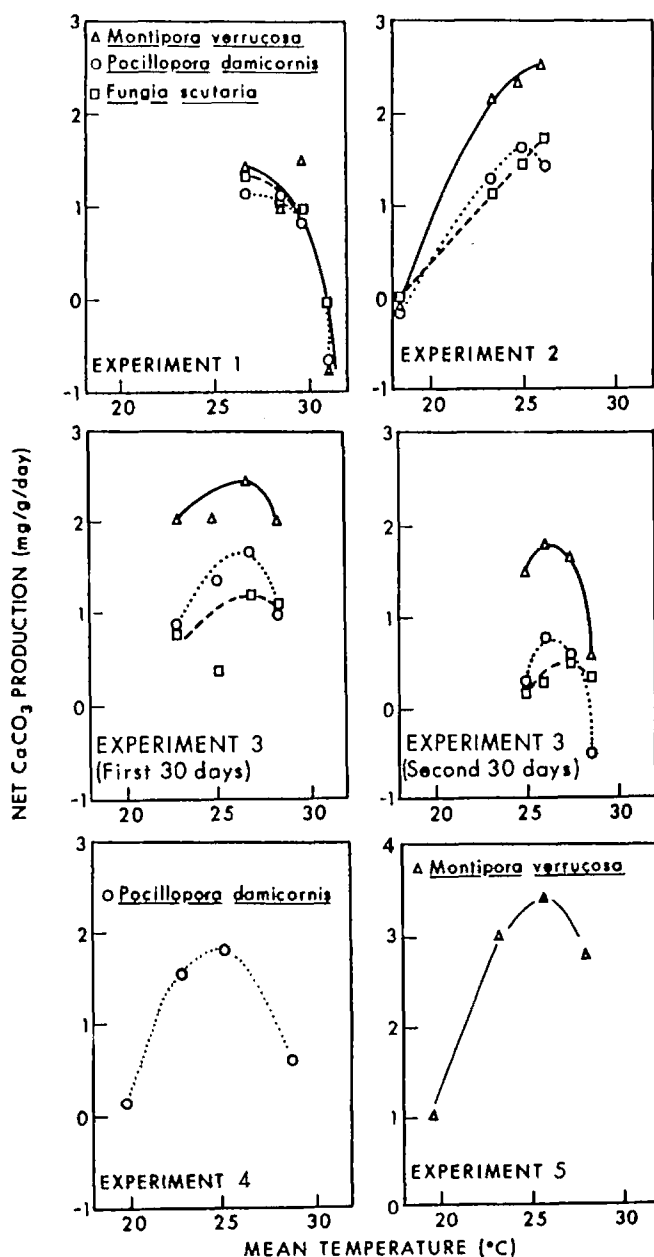


Fig. 2. Rate of change in total skeletal mass of corals for the various experiments

natural incident solar radiation. Often shaded portions on the underside of the colony survived. Apparently coral tissues can better withstand temperature extremes at reduced light intensities. During Experiments 4-5 no mortality occurred because temperature was kept in the 20° to 30°C range.

Seven-day exposure to 30°-32°C (Experiment 6) caused loss of pigment but no mortality in mature *Pocillopora damicornis* colonies. Approximately half of the newly-settled *P. damicornis* were killed during the 7 day heat stress period, and the remainder showed pigment loss. Thirty-two days after the start of the experiment nearly all of the new settlements were dead. All of the mature colonies recovered.

Skeletal Growth

Rate of change of total coral skeletal mass ($\text{mg g}^{-1} \text{ day}^{-1}$) in each aquaria as a function of temperature is presented in

Fig. 2. A calcification optimum near 26°C is indicated for all three species. *Montipora verrucosa* appears to calcify most rapidly and *Fungia scutaria* least rapidly. During the first 30 days of Experiment 3, restricted water flow to the ambient tank (25°C), caused stagnation and reduction in the calcification rate. Fig. 2 shows a reduction in calcification rates of all three species in the 25°C treatment during this period relative to the other treatments.

Experiments 4 and 5 suggest an optimum growth temperature of approximately 26°C. Experimental temperatures extended across a range from lower to upper sublethal. Bleaching occurred at the upper sublethal (28.1°C) treatment. Skeletal growth was reduced at the sublethal (20.0° and 28.1°C) treatments and greatest at the temperature that most closely approached optimum (25.9°C).

Similar differences in growth of *Montipora verrucosa* with temperature were found in Experiment 5 as determined previously (Fig. 2). Moreover, the larger sample size used in this experiment also enabled an analysis for statistical differences in growth among temperature treatments. The size range of corals used in Experiment 5 was 2 to 30 g buoyant weight. Size variation was found to influence comparisons if growth was expressed as percent increase of initial weight. It has been established that radial growth in hemispherically shaped corals of the size range used in these experiments is largely independent of initial size (Knutson *et al.*, 1972; Buddemeier *et al.*, 1974) and proportional to the cube root of the weight (Kawaguti, 1941; Maragos, 1972). Therefore, it is possible to reduce the influence of initial size by expressing growth as the difference between the cube roots of the initial and final weights ($\Delta\sqrt[3]{\text{weight}}$). This transformation produces a number that is proportional to the thickness of skeletal material added to the coral during the growth period, and reduces the complicating effect of differences in initial weight. A complete review of the coral growth literature (methods, assumptions, results) is contained in Buddemeier and Kinzie (1976).

A comparison of transformed growth rates for corals at each temperature is given in Table 3. Differences between pairs of means were tested by calculating a least significant difference, D ($P < 0.05$), using a pooled standard error of treatment means (Snedecor and Cochran, 1967, pp 272-273). Growth in the 25.8°C treatment was significantly greater than at 28.1°C. Growth at 19.8°C was significantly less than at all other tempera-

tures. The results indicate that growth was curvilinearly related to temperature, with the 25.8°C treatment most closely approaching optimum.

Recovery of Bleached Corals

The control groups of 10 normal corals of each species returned to the field showed no changes during the experiment. All bleached specimens partially recovered zooxanthellar pigment within 1 month, with full recovery occurring within 2 months (Table 4). The topmost surface of the coral exposed to full natural light intensity during thermal stressing appeared to be less capable of recovery. In general, tissue on the upper surfaces or tips of the colonies died while tissue on the undersides and shaded parts of the head survived. Rate of pigment recovery appeared to be the same for all three species.

Corals damaged by high temperature can recover if returned to optimal conditions, but mortality of bleached specimens is high during the recovery period. Ability of the species to recover from high temperature damage appears to follow the same relative order as initial thermal resistance. In contrast, corals damaged by low temperatures (18° to 19°C) were not able to recover when temperature was returned to normal.

Discussion

The upper thermal limits established for *Pocillopora damicornis*, *Montipora verrucosa* and *Fungia scutaria* by this study are in general agreement with previously reported values for Hawaiian corals. Edmondson (1928) reported slightly higher values (32° to 33°C) for these species, but his determinations were based on short-term exposures. A more recent field study (Jokiel and Coles, 1974) suggested an upper lethal limit of 31° to 32°C for prolonged exposures.

Differences in reported values for upper lethal temperature have been shown to be related to the length of exposure to elevated temperature, and subtropical corals appear to have an upper lethal limit that is 2 C° lower than their tropical equivalents (Coles *et al.*, 1976). Apparently Hawaiian corals have a restricted capacity to adapt to tropical temperatures. Corals maintained near 30°C for 2 months in this study eventually lost symbiont pigment and showed increased mortality, rather than undergoing acclimation. This also has been observed under field conditions (Coles, 1975). The same species apparently pros-

per in parts of the tropical Indo-Pacific, where ambient natural water temperatures remain near 30°C for prolonged periods (Coles *et al.*, 1976). Possibly the process of adaptation in corals requires many generations. Alternatively, a different genetic strain of alga symbiont might inhabit the tropical representatives of the various coral species.

The prolonged (1 month) lower lethal limit of approximately 18°C determined in this study agrees with estimates derived from zoogeographic distribution (Vaughan and Wells, 1943; Wells, 1957). Corals can survive low temperatures for short periods, leading Mayer (1914, 1918) and Edmondson (1928) to conclude that corals are most sensitive to elevations in temperature than to similar temperature reductions. The same conclusion could be drawn from our results if exposure time were 1 week or less. Lengthening the exposure time to several weeks led to complete mortality (Fig. 1) and the opposite conclusion.

Both experiments resulted in similar mortalities for the 30-day period. However, virtually opposite patterns of mortality with time are shown by the two treatments (Fig. 1). At the upper lethal limit, mortality rate was high in the first few days but decreased with time. Conversely, at the lower lethal limit mortality rate was initially low but increased with time. At the upper lethal temperature treatment the heavy damage (loss of pigment and patches of tissue) that preceded mortality was apparent within the first 24 h of treatment. At the lower lethal limit visible changes did not occur until after several days but then accelerated with time. Although the corals initially appeared to be resistant to the lower temperatures, they apparently had little or no ability to acclimate.

We did not observe any relationship between initial size and thermal resistance over a size range of from approximately 2 to 200 g wet weight, but new coral settlements appeared to be more sensitive to high temperatures than mature colonies. New settlements are probably the most sensitive life stage of the coral, since the larvae are known to tolerate higher temperatures than adults (Edmondson, 1946).

On the basis of zoogeographic distributions, reef corals have been assumed to have optimum growth at 25° to 29°C (Vaughan and Wells, 1943). Laboratory investigations (Yamazato, 1970; Cox, 1971) and field studies (Shinn, 1966; Maragos, 1972; Glynn and Stewart, 1973) have indicated that coral calcification in the subtropics increases with in-

creasing temperature up to and in some cases beyond 26° to 28°C. Several studies (Shinn, 1966; Yamazato, 1970; Clausen, 1971) indicate a skeletal growth decrease above 28°C. Clausen and Roth (1975) found a 27°C growth optimum for Hawaiian *Pocillopora damicornis* and a 31°C optimum for the same species at Enewetak Atoll, Marshall Islands, where ambient water temperatures are higher. An optimum growth temperature of approximately 26°C was indicated by the long-term temperature responses of the three Hawaiian corals in the present study. Also, statistically significant reductions in growth above and below this optimum were demonstrated. The temperature optimum lies closer to the upper lethal temperature limit than it does to the lower lethal limit.

The potential effects of climatic temperature changes on Hawaiian corals can be determined by comparing mortality and growth responses (Figs. 1 and 2) with seasonal water temperature curves for Kaneohe Bay, Hawaii (Fig. 3) and adjacent areas. In Hawaii, water temperatures fall below the optimal 26° to 27°C range for 7 months of the year. It is apparent that any downward shift of the annual water temperature curve will decrease growth during all 12 months of the year. A slight increase in temperature (less than 1 C°) would increase growth in nearly all months for a net annual increase in calcification. Even an increase as large as 2 C° would not decrease net annual growth, because a shift that raised the summer temperature above the optimum would be compensated for by corresponding increases in the cooler months.

Near the northwest terminus of the Hawaiian chain lies Midway, one of the islands marking the northern limit of reef development in the Pacific Ocean. The coral fauna of Midway is closely related to that of Hawaii, and during the winter months these corals experience water temperatures that closely approach the lethal limit (18°C) of their Hawaiian counterparts (Fig. 3). The nearest island to the south of Hawaii is Johnston Island. Although still considered to be in the Hawaii subprovince, Johnston Island differs significantly in that the genus *Acropora* is present, possibly because the winter temperatures do not fall below 25°C. Temperatures at Johnston Island are near optimal throughout the year, a possible explanation for the very high coral coverage found at this location (Brock *et al.*, 1965). The Johnston Island, Hawaii and Midway Island temperature curves differ little (less than 1 C°) in the summer months. The main differences occur in the length

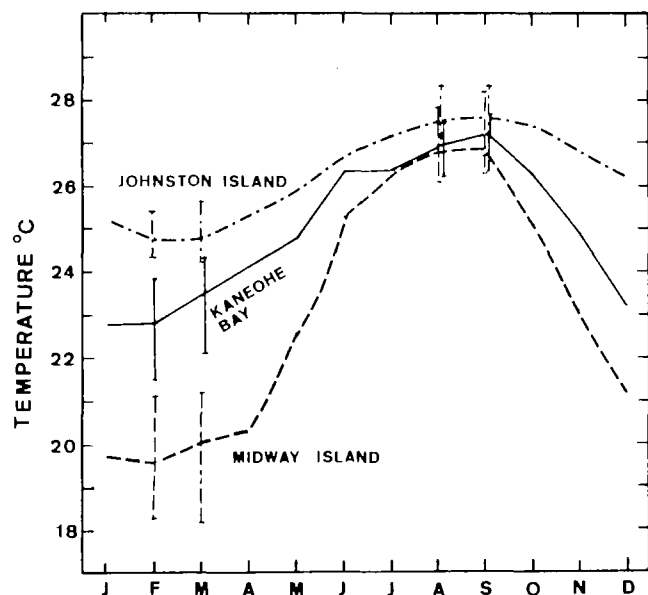


Fig. 3. Long-term mean monthly ocean temperatures for Kaneohe Bay, Hawaii compared to northernmost (Midway) and southernmost (Johnston) islands of the Hawaiian coral subprovince. Vertical bars indicate range of mean monthly values reported between 1960 and 1969. (Data from Anonymous, 1970)

and severity of cold water conditions in the winter. It appears that winter temperatures are of crucial importance to the development of reef coral communities in the Hawaiian region.

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